

CONDITIONED TASTE AVERSIONS AND LATENT INHIBITION IN EGYPTIAN SPINY MICE AND LONG-EVANS RATS¹

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Summary.—The acquisition and extinction of a conditioned taste aversion in Egyptian spiny mice and Long-Evans rats was compared during 20 posttest sessions using a cross-over design and double-blind control procedures. Spiny mice preexposed to a sucrose CS demonstrated more latent inhibition and a faster rate of extinction than did Long-Evans rats preexposed to the same CS. Preference indices did not differ between control animals or as a function of gender. The present results are the first report of the effects of latent inhibition on learning taste aversion in Egyptian spiny mice.

In prior studies we described noteworthy differences in the ingestive behavior of Long-Evans rats and Egyptian spiny mice (Kolodiy, Brosvic, Bailey, Hawley, Pak, & Ostrich, 1993; Kolodiy, Brosvic, Pak, & Loeffler, 1993; Loeffler & Brosvic, 1993). In these studies the ingestive behavior of spiny mice was remarkably different from that of Long-Evans rats, with gustatory cues appearing to be less salient to spiny mice. These differences were observed during need-free conditions as well as during acute and chronic states of water- and sodium-deprivation and raise interesting questions about the role(s) gustatory cues might play in the spiny mouse's acquisition of a conditioned taste aversion.

To our knowledge there has been only one prior report on the learning of taste aversions involving spiny mice (Etscorn, 1977), with a robust aversion observed after a single exposure to sucrose paired with lithium chloride. The results of pilot work conducted in our laboratory support this prior demonstration. We also observed that the taste aversion of a small number of spiny mice was reduced by preexposure to a sucrose CS substantially more than that observed for Long-Evans rats. In the present study we undertook a more extensive examination of learned taste aversions in Egyptian spiny mice and Long-Evans rats using double-blind control procedures and a cross-over design.

METHOD

Thirty female and 10 male in-bred Egyptian adult spiny mice (*Acomys*

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cahirinus) and 20 female and 20 male in-bred Long-Evans rats served as subjects. The animals were housed in polystyrene cages in a temperature- and humidity-controlled vivarium with Purina Lab Chow available *ad libitum*. All animal maintenance and testing protocols were approved by the Rider University Institutional Animal Care and Use Committee.

Fluid (triple-deionized water) consumption was measured for each animal in five baseline sessions. Two groups of rats and two groups of spiny mice were matched on weight and fluid consumption. A .25-M solution of sucrose was used as the conditional stimulus (CS). Conditioning was conducted in two phases, each consisting of one conditioning day (Test Day 0) and 10 Posttest Days. On Day 0 of Phase 1, one group of rats and one group of spiny mice received intraperitoneal injections of .15 M lithium chloride (20 mg/kg) after 30 min. of preexposure to .25 M sucrose. Control animals received intraperitoneal injections of 0.9% saline (20 mg/kg) following 30 min. of preexposure to triple-deionized water. During Posttest Days 1–10 each animal was then presented two bottles, one containing triple-deionized water and the other containing .25 M sucrose. On Day 0 of Phase 2, a cross-over design was initiated with the former control animals injected with LiCl paired with sucrose and the former experimental animals injected with saline paired with triple-deionized water. Testing then continued according to the procedures described.

A preference index was calculated for each test session using the formula described by Etscorn (1977): [amount of sucrose CS consumed]/[amount of sucrose CS consumed+amount of triple-deionized water consumed] $\times$ 100. Accordingly, a preference index value of 50 represents an equal preference for sucrose and water whereas a value of 75 indicates a preference for sucrose and a value of 25 the rejection of sucrose. All data were collected according to double-blind control procedures and analyzed using a mixed model analysis of variance and Scheffé comparisons.

RESULTS

Significant main effects and interactions were observed for species, phase, and test day (all $F > 18.93$, all $p < .05$) and median preference indices are presented for the purposes of comparison. In the absence of preexposure to the CS (Phase 1) no between-species (mouse $Mdn = 8$; rat $Mdn = 4$) differences in preference indices were observed at Posttest Day 1. On Test Day 2 the preference indices of spiny mice were significantly greater than those observed for Long-Evans rats, and this difference was also observed on Posttest Days 3–8. The taste aversion in spiny mice extinguished in significantly fewer test sessions (mice by Posttest Day 7, $Mdn = 77$; rats by Posttest Day 9, $Mdn = 81$). In Phase 2, the preference indices were significantly higher on Posttest Day 1 for spiny mice ($Mdn = 30$) than for rats

($Mdn=16$) and this difference was again observed on Posttest Days 3–6. The Phase 2 taste aversion for spiny mice extinguished in significantly fewer test sessions (mice by Posttest Day 5, $Mdn=82$; rats by Posttest Day 7, $Mdn=78$). No differences in preference indices or rates of extinction were observed between the control groups or as a function of gender at any point during the study.

DISCUSSION

The learning of a taste aversion by desert strains of rodent has historically been difficult to establish and was first reported for the Egyptian spiny mouse by Etscorn (1977). In that study a novel CS (sucrose) was used and the preference indices for the present study's Phase 1 (Posttest Day 1) experimental animals are similar to those reported by Etscorn (1977). As the Etscorn (1977) values were made in the absence of preexposure to the sucrose CS, it appears that novel gustatory cues are as salient to Egyptian spiny mice as they are to Long-Evans rats. In contrast, the preference indices observed for spiny mice preexposed to sucrose were significantly higher than those for Long-Evans rats preexposed to sucrose, suggesting that latent inhibition was more influential on spiny mice. The present results were observed after 10 preexposures to sucrose and some parameters were examined and the findings replicated in additional studies in which the effects of varying degrees of preexposure to sucrose (0, 3, 9, or 27 sessions) on learning taste aversion were examined. In those studies more latent inhibition and a faster rate of extinction were observed for the spiny mice after three or nine but not after 27 preexposures to the sucrose CS. The results of these latter studies suggest important limitations on the over-all influences of latent inhibition on the ingestive behavior of spiny mice in particular and on aversive conditioning in general.

It should be noted that the procedures used in the present study were similar to those described by Etscorn (1977) and thus do not permit control of potential differences in the amount of preexposure to the sucrose conditional stimulus. The median amount of fluid consumed by spiny mice did not differ between the conditioning ($Mdn=3.1$ ml) and the final posttest days ($Mdn=3.4$ ml). Fluid consumption by the rats also did not differ between the conditioning ($Mdn=35.3$ ml) and the final posttest ($Mdn=33.2$ ml) days. However, when adjusted for the substantial differences in weight between the spiny mice ($Mdn=58$ g) and rats ($Mdn=224$ g), the rats consumed significantly more fluid at each measurement interval. This latter outcome is consistent with our prior reports of substantial differences between the ingestive behavior of spiny mice and Long-Evans rats (Kolodiy, Brosvic, Bailey, Hawley, Pak, & Ostrich, 1993; Kolodiy, Brosvic, Pak, & Loeffler, 1993; Loeffler & Brosvic, 1993). Additional studies are needed before

the influences of these factors on the outcomes of the present study can be determined. Given the precocial nature of the spiny mouse (Brunjes, 1990) and the recent demonstration of an effect of preexposure to a conditional stimulus (using a light-tone combination and footshock) in immature Sprague-Dawley rats (Kraemer & Randall, 1992), it would also be of interest to examine developmental trends for aversive learning in the precocial spiny mouse.

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